Will Information and Incentive Affect Traveler's Day-to-Day Departure Time Decisions? - An Empirical Study of Decision Making Evolution Process

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Abstract

The uncertainty and unpredictability of a transportation network roots in the dynamics of individual travel behavior, which can be revised consciously or repeated habitually depending upon the reality and personality. In this paper, we propose to study the day-to-day departure time choice behavior of the travelers, using real observation data collected from a smartphone app, "Metropia". Influenced by the information and incentives provided in the app and the comparison with the experience gained from the last trip, a transformation process of traveler's day-to-day experience on departure time from an existing habit to a new one is analyzed in this study. The analysis result in a binary choice model for the shift of departure time for each repeated morning commute trip comparing with the last one, which proves that users' experience in app engagement, previous travel time saving, habitual travel time, incentives, and commute flexibility are able to trigger day-to-day behavioral change for their morning home-to work commutes. The findings of this research provide insights on the users' adaption to a new traffic information service along with incentives, and corresponding behavior changes over a transition period. The outcomes suggest ways to improve ICT services and incentive scheme, as well as the transportation operation and demand management.

Keywords*: Departure Time Choice, Travel Behavior evolution, Smartphone Based Data Collection, Incentive rewards, Information and Communication Technology (ICT)*

1. Introduction

The uncertainty and unpredictability of a transportation network roots in the dynamics of individual travel behavior. The daily decisions of a repeated trip, such as route choice and departure time choice, can be revised consciously or repeated habitually depending upon the reality and personality. Understanding the day-to-day travel behavior and decision-making is one of the major challenges in travel behavior research. These behavioral dynamics and randomness, especially for departure time in this study, are influenced by the personality and habit of travelers, the real-time traffic information, and the up-to-date perceptions and experiences. With the updating process from the past trips, one would suspect the travel behavior may reach certain stability in the long run, until significant changes in the traffic or other exterior stimulus breaks the equilibrium.

Traffic information is one of the factors highly associated with departure time decision making. With the use of information & communication technology (ICT) emerges among the population, more and more traffic information from various channels are collected and shared with public and social networks instantaneously. The real-time travel time and accidental information can help travelers plan ahead and adjust their departure times based on personal preferences and needs. For example, a commuter can leave later to avoid sitting in the queue caused by an accident, or a shopping trip can be canceled or rescheduled because of a foreseen delay. As the new source of information and new information channel appears, there is also a learning cycle that travelers adopt the information and cumulate experiences of the information, and then gradually adapt to the fresh information with individual judgments.

Incentive, which could take forms of monetized lottery/rewards, free transit ticket, travel feedback (such as gas saved), or environmental contributions, is increasingly applied together with the state-of-art ICT in making a fundamental paradigm shift from the traditional passive travel information provision to an active traffic demand management tool, by influencing people's need and intention to travel as well as their associated travel pattern. With the impact of incentives, the information has more meaningful suggestions for the traveler and helps the decision making process about what time to leave, what mode to use, which route to take, or even to travel or not.

Considering the above-mentioned two major motivations, a traveler’s day-to-day experience on departure time is essentially a transformation process from an existing habit to a new habit, with repeated trials and deliberations. The aim of this paper is to analyze the day-to-day departure time choice behavior of the travelers, and explain how the commuters change their departure time gradually with a new service of traffic information along with incentives to form a new habit. The study is based on true observation data collected from a smartphone app, “Metropia”, with the hope of filling the gap in departure time behavior study by understanding the impact of incentives and the engagement with ICT services in a real-world setting on the day-to-day evolution of departure time decision. The findings of this research could potentially shred lights on how to better management traffic demand with improved

ICT services, and improve the travel demand forecast mechanism in both short and long term.

The rest of this paper is organized as follows: Section 2 reviews the relevant past literature on departure time decision making process and day-to-day behavior change; Section 3 presents the data collected for the analysis. Section 4 describes the departure time modeling process and the results. Section 5 concludes this research and discusses future research directions.

2. Literature review

As one of the fundamental travel behavior dimensions, departure time choice is highly associate with the travel demand, especially for the peak hours. There have been plenty of literatures on departure time choice modelling. For example, Mahmassani presented a modeling framework to investigate the dynamic properties of a system of commuters located along a highway facility [1, 2]. Bhat built a continuous-time model of departure time choice for urban shopping trips [3], and a mixed multinomial logit structure for home-based social–recreational trips [4]. Mannering investigated the determinants of commuter flexibility in changing routes and departure times for the morning trip to work [5, 6]. Szeto developed a cell-based formulation for the simultaneous route and departure time choice problem with elastic demands [7]. Jou studied the dynamic commuter departure time choice under uncertainty [8]. Chang examined two heuristic rules for describing commuters' predictions of travel time as well as the adjustments of departure time in response to unacceptable arrivals in their daily commute [9]. Other researches on departure time choice can also be found in [10-12].

Over the last few decades, researchers start to realize that travel behaviour decisions are both dynamic and stochastic and are made by the travellers on a day-to-day, trip-by-trip basis. The real-time information with the widely accepted ICT intensify the uncertainty in the day-to-day decision, as dynamics in the traffic situation and the instantaneous update of the traffic information increase the needs in modelling and analysing the travel behaviour in a dynamic manner. Many studies tried to model the travel behaviour in route choice and perception in travel time [13-16]. For example, MITTENS (Messaging Infrastructure for Travel Time Estimates to a Network of Signs) in San Francisco provides real-time freeway and CalTran travel-times information to motorists in order to induce en-route mode shifts [17]. In San Francisco area, Predict-a-Trip^{SM} was built based on the popular 511 Driving Times^{SM} service, which can be accessed by travelers via phone, web or freeway message signs. It is helpful for motorists to obtain route information not only for planning trips, but also for them to consider taking public transit instead of driving, or to use the ride-matching tool for prospective carpools [18].

As opposed to road pricing that has been proved efficient yet controversial [19], the effectiveness of incentives to reinforce a desirable behavior is supported by a large volume of empirical evidence [20, 21]. However, the implementation and relevant research on incentives in the transportation area has a relatively short history. During an 8-day temporary freeway closure in Osaka, Japan, some 20s travellers were offered a one-month free bus ticket and observed to have used the bus more

frequently after the intervention [22]. According to a longitudinal study in Germany, which offered the incentive of a pre-paid bus ticket, the bus usage has increased by 35% and trips finished by car decreased by 13% [23]. In Melbourne, Australia, an early bird ticket program was proposed to alleviate the rail overcrowding issue during the peak hours and proved to be effective [24]. A recently developed smartphone app-based travel feedbacks help reduce car use by 33% in the University population [25]. Tripzoom, an app from Germany, builds in the ability for participants to set goals [26]. In the Netherlands, rewards are well-recognized as the extrinsic motivation for discouraging rush-hour driving, in which experiment a smartphone called 'Yeti' that worths 500€ is provided as reward, and an individual can gains some credits ranging from 3€ to 7€ depending on the time of a day [19, 27]. In the US, Metropia, a new technology company, also explored to use a combination of incentives and information to change people's behavior of multiple dimensions, such as improving people's driving behavior for the purpose of reducing driving crashes [28].

From those literatures, we can tell while departure time choice problem has been extensively studied, almost all prior studies approached this topic either via mathematical formulation or simulation and came with various assumptions. The departure time choice study based on the continuous observation of real traveller's day-to-day engagement with ICT services in a real-world setting, influenced by the dynamic traffic information and incentives offered, and focusing on the analysis of day-to-day departure time choice behavior and how the commuters change their departure time gradually with a new service of traffic information along with incentives to form a new habit, is rather limited, which leaves the gap for this research to fill in.

3. Data

The data used in this study is collected from a smartphone app "Metropia", which uses traffic prediction and coordinating technologies combined with user rewards to incentivize drivers so as to balance the traffic load on the network, and reduce traffic congestion. Drivers use the app to plan a trip, and the system provides travel time prediction for every 15 minutes and varying levels of incentives which are based on the traffic condition with the goal of attracting users to the less congested departure times. The reward points are determined based on the departure time and congestion levels of the road. If the departure time and route selected by user is heavily congested, the points will be lower, and if the users are willing to leave during the non-peak hours, they can get more points. Once user decides on the time and route to travel, the system will verify if the user indeed leave at the time promised and follow the desired route, and if yes, the trip is validated and the points will be added to the user's account. The user can keep using the app to accumulate the points. The points carry monetary value, and can be used to redeem freebies such as Amazon and Starbucks gift cards. With the app, the trip related record is collected, including origin, destination, departure time, distance, the actual experienced travel time, the system suggested travel time, incentive points received, user saved favorite location, etc. As the user continues to make trips with the app after they sign up, the system collects and stores the trip information for all the trips and the interactions between

user and app. Screenshots from the app are presented in the following figure to describe how the app works. Additional details of the system including the travel time prediction mechanism and accuracy, reward points can be found in [29, 30].

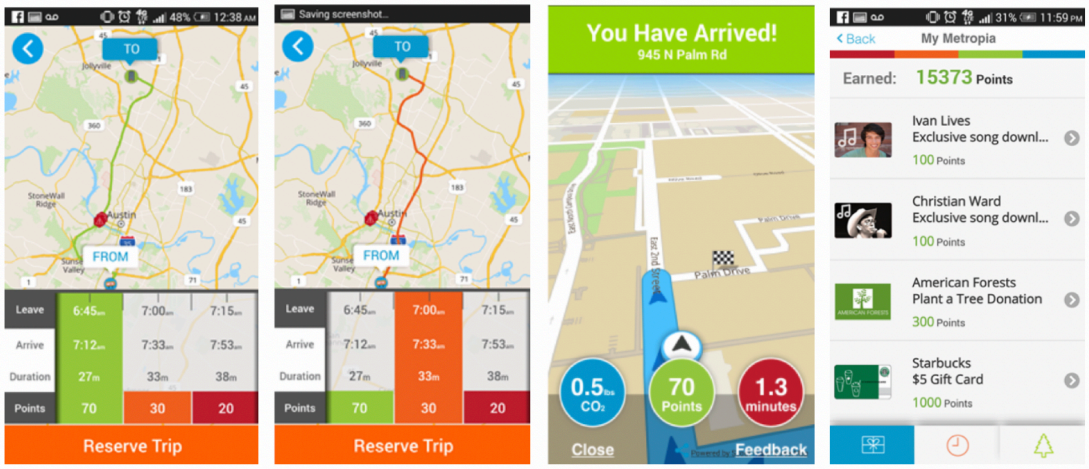


Figure 1. How System Works

In this paper, we randomly selected 157 frequent users in Austin who signed up during May to Nov, 2015 and completed more than 50 trips during this period. The rational is that if a user has finished over 50 trips, he/she should have a good idea about what's in the app, and his/her continuous travel behavior observed from the first day using the app is sufficient enough to be studied.

In total, we collected 11,709 valid trips during this period. We firstly divide all travelers to commuter and non-commuter, based on their trips patterns between frequent locations, especially home and work. And the trips between frequent OD pairs are separated to a total of four study groups, namely 1) commuter home to work trips, 2) commuter work to home trips, 3) commuter other trips, and 4) non-commuter trips. The reason is commuters and non-commuters come with different constraints (in particular, those constrained by work hours) and subsequently different travel behavior. For example, for the same grocery shopping trip purpose, commuters are more likely to do so after work (say after 5pm) with some but limited departure time flexibilities, while non-commuters can do so at any time in the day and will have more departure time flexibility. Based on such rational, we feel it's the best to divide users by their work status in addition to trip purposes. The data characteristics for each group are listed in Table 1 below. In total there are 157 distinct users, including 132 commuters and 25 non-commuters. Among those 132 commuters, 73 users' home to work have been observed for multiple times (and thus only 73 users are listed in row 2, column 2). Those users overlap, meaning there're some users have frequently observed home to work trips, and/or work to home trips, and/or other trips, so column 2 does not add up to be 157.

Table 1 sample size for each study group and properties

	number of user	number of OD pairs	number of trips	as first trip in the day	as last trip in the day	median of stay duration before next trip in the same day
Commuter home to work trip	73	74	3604	3210 (89.1%)	588 (16.3%)	7.6 hour
Commuter work to home trip	62	63	2644	279 (10.5%)	1910 (72.2%)	1.3 hour
Commuter other trip	42	125	2904	823 (28.3%)	847 (29.2%)	1.2 hour
Non-commuter trip	25	96	2557	1031 (40.3%)	859 (33.6%)	1.6 hour

As Table 1 shows, home to work trips are mostly the first trip in the day and work to home trips are mostly the last trip in the day for commuters. The median value of stay duration at work is 7.6 hour, which is close to reality and is the longest among the four groups. For those work to home trips that are not the last trip in the day, the stay time is about 1 hour on average, before activities at night. For other trips of commuters (such as shopping, recreation) and no-commuters, there is no strong indication that the trips should be the first or the last trip in the day, and the stay duration at the destination is about 1.2 and 1.6 hours on average.

If we visualize the OD pairs in spatial and temporal features for a typical traveler as example in Figure 2 (the x and y axes are longitude and latitude for the geospatial location, and z axis is the timeline that spans multiple days), we can find out that commuters' trips are quite regular and are recurring frequently. And trips to other locations account for only a small percentage of the total trips and are randomly distributed. In contrast, example of OD pairs for a typical non-commuter visualized in Figure 3 does not reflect a strong pattern in the recurring OD pairs. The frequent trips can be shopping or escorting kids, but in general the pattern is not as clear as commuters. After visualizing and manually inspecting the trip pattern for multiple travelers, similar pattern as Figure 2 and Figure 3 can be found and we generate this argument for the rest of the samples.

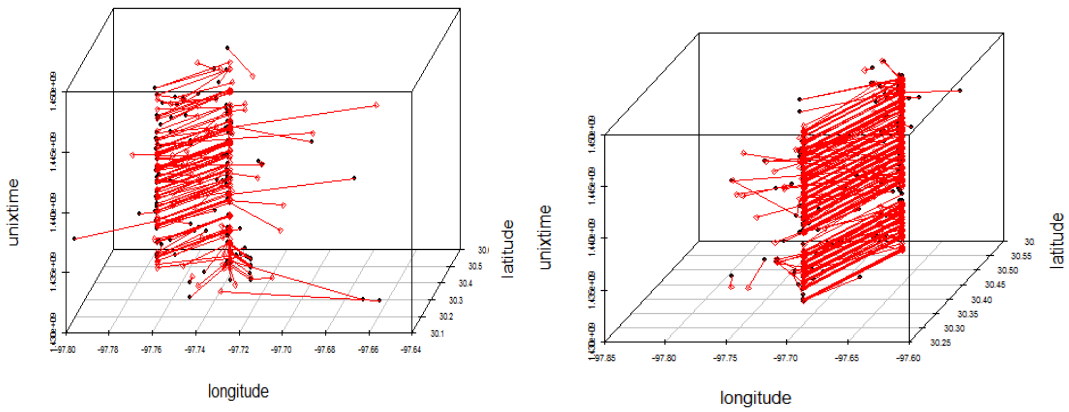


Figure 2. OD pattern for commuters

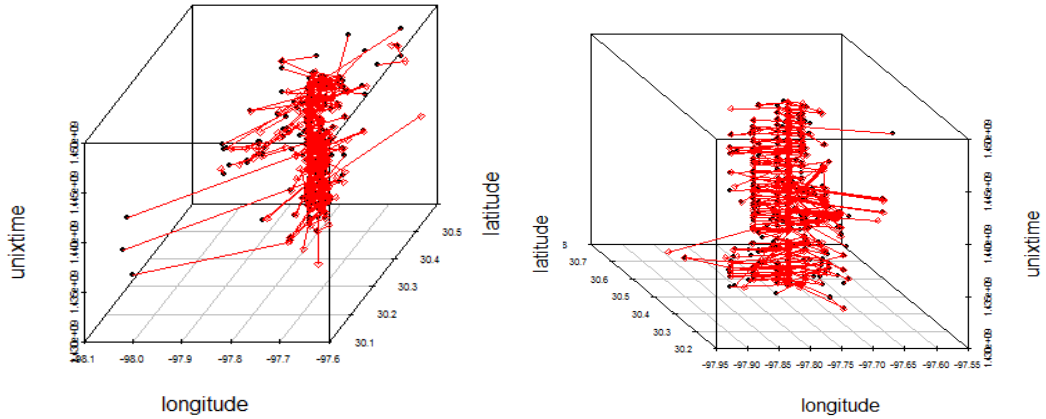


Figure 3. OD pattern for non-commuters

4. Behavioral Analysis

Based on the continuous observing user's travel behavior and departure time choice, we propose to answer the following questions.

1. Will information and incentives provided affect a driver's departure time choice, and if so, will the new habitual departure time form and stabilize as time goes?
2. What are the factors that effectively influence departure time decision, especially, how the day-to-day decision evolves because of the experience with the traffic information and incentives from the ICT platform?

The procedure to analyze the data is discussed in following two sub-sections. First, we study the departure time choice evolution pattern, and if they are different for different type of trip purposes. The next step is to understand among all the factors communicated between the traveler and the app, what effectively triggers the day-to-day decision changes. Home to work trips are used as the sample for the model estimation for this step.

4.1 Departure time evolution analysis

In this section, we plan to investigate for the multiple trips between the same OD pair with the same trip purpose. Start from visualizing departure time within each trip purpose, Figure 4 illustrates the departure time when the user travel with the app for the first 20 times between the specific popular OD pairs, with X as the number of time the user travels and Y as the departure time for each travel. It can be found out that home to work trips are more concentrated from 6am to 10am and the lines are mostly flat. For the frequent fluctuations in the departure time between 8am and 2pm or 7 pm, we suspect this type of users has a regular work shift schedule, such as the work shift schedule for nurses. Work-to-home trips have a wider bandwidth and they mostly happen between 3 pm and 7pm. The departure time for other trips for commuters and non-commuters are not showing a clear stable pattern.

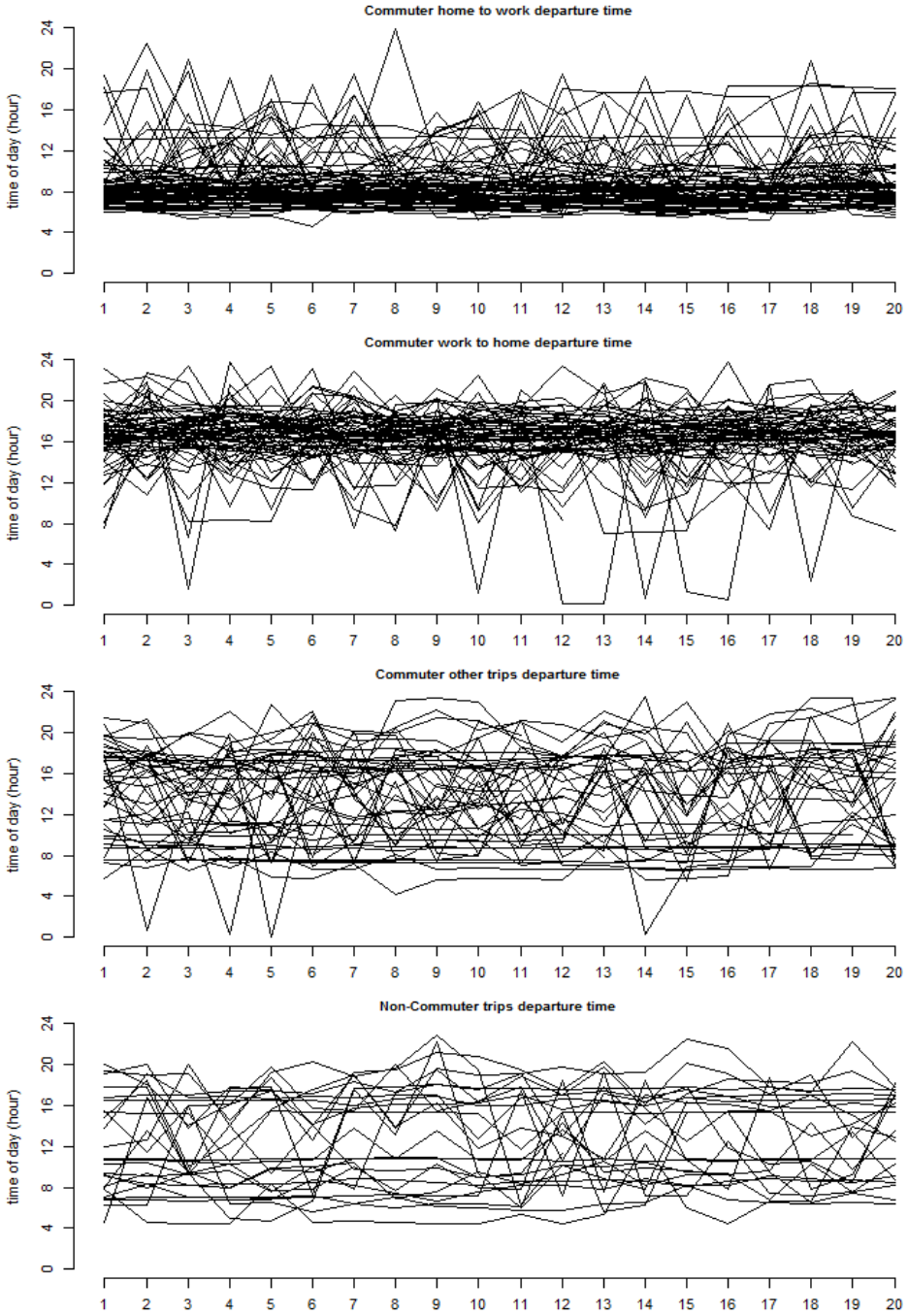


Figure 4. Departure time for the first 20 times between the specific popular OD pairs

The shift in departure time is defined as an indicator of how many minutes the departure time deviates between current trip and the previous trip with the same OD. Comparing the home to work trips and non-commute trips, as shown in Figure 5, the standard deviation value of the shift in the departure time fluctuated with a decreasing trend as the trip gets repeated, indicating a convergence pattern in the departure time to a stable state. The number of repeated observations of non-

commuter trips (shown as +) in Figure 5 is lower than home to work trips (shown as triangle), as there are fewer observations of the non-commuter trips repeated more than 50 times. The decreasing rate is slightly faster for home-to-work trips than the non-commuter trips, meaning both trip categories will converge, but home-to-work trips converges faster than the non-commuter trip, most likely due to the fact that the later has more flexibility in their departure time.

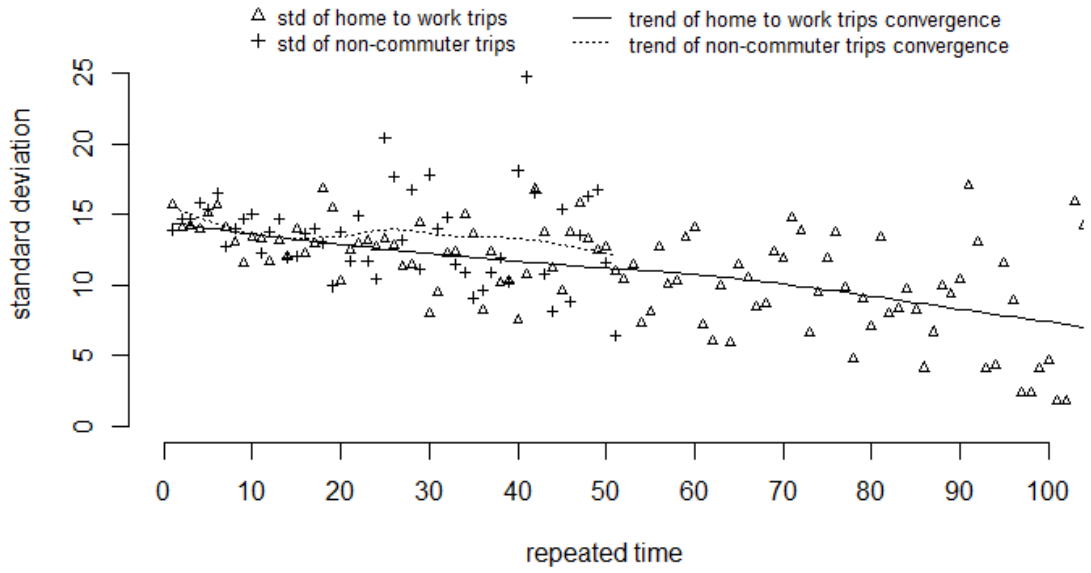


Figure 5. Trend of standard deviation representing convergence

Among the above home to work trips, we classify the trips as two groups: *non-flexible schedule* and *flexible schedule*. The trips during morning period (4am to 10am) between the same OD pair are selected and sorted by the time. As shown in Figure 6, 9 users (about 12%) have a relatively strict working schedule, as the maximum difference in their departure time is no larger than 30 minutes. The rest 64 users (88%) are having a relatively flexible schedule. Therefore, a major portion of the commuters has the potential to shift their departure times, and can be the target users to encourage behavioral changes and reduce peak hour traffic congestion. Here departure time is selected to determine user's schedule flexibility, as opposed to the arrival time, as we believe departure time is user's "decision" to make which he/she has control and can change if not desired, while arrival time is the "result", and thus the departure time is more suitable for the modeling process. The impact of this factor to trigger the behavior change will be studied continuously in the next section.

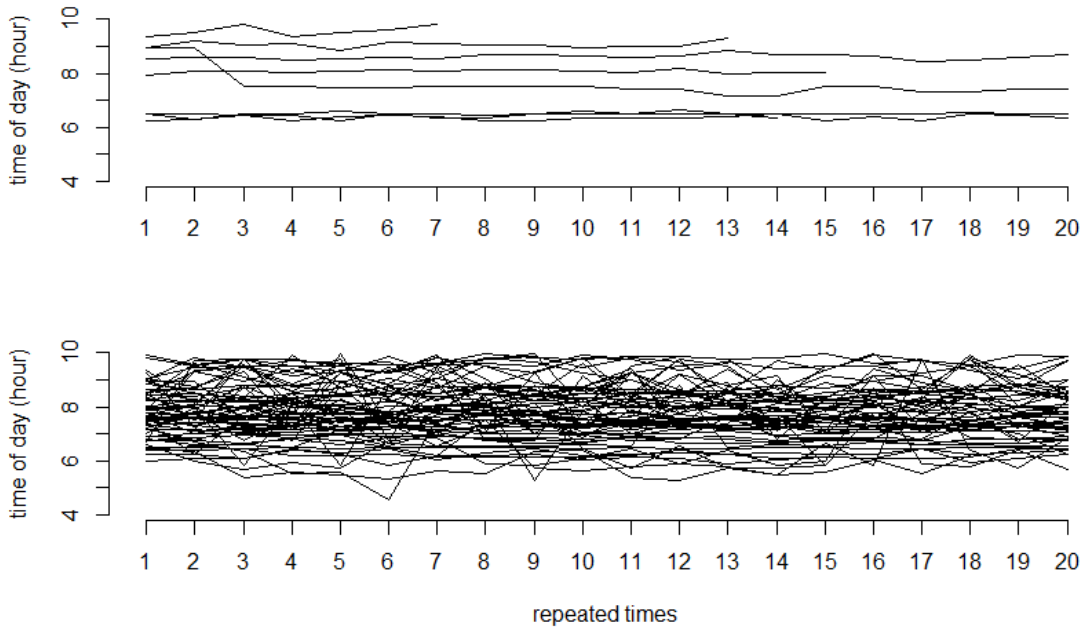


Figure 6. Peak hour home-to-work trip departure time evolution

4.2 Traveler behavior evolving process modeling

The reasons that the users change their departure times or stick to the original departure times include the experience and accumulating knowledge about the traffic conditions, as well as the incentives provided by the smartphone app. Before the modeling process, the hypothesis to be tested is as follows:

1. The user will make decision based on their past experience and the current available information.
2. The decision of changing departure time or not is influenced by factors including travel time saving from the previous trip, i.e., whether actual travel time was less than the estimated travel time for the last trip, and if the suggested departure time corresponds to a shorter travel time than the last trip.
3. The rewarding scheme can also influence the decision-making process.

To make this manuscript compact, the dataset used in the modeling is morning peak hour home-to-work trips for the commuters. This is also because 1) there are more long-term observations for morning commutes among the four types of trip purpose, and 2) morning home-to-work trips shows a stronger pattern to converge to stability in the departure time choice, and as they contribute significantly to the peak hour congestion, the study of home-to-work trip has its significance. The variables used in this model are described in Table 2:

Table 2 Descriptions of the explanatory variables

Variable	Description	mean	std
prev_est_tt	Estimated travel time of the previous trip	22.92	9.931
prev_act_tt	Actual travel time of the previous trip	20.64	10.33
prev_exp	Difference between the previous estimation and true travel time	2.28	6.28
current_est_tt	Estimated travel time for the departure time that user chooses to leave	22.85	9.88
prev_tt_saving	Travel time saving of the previous trip	3.59	4.30
prev_Mpoint	Points earned from the previous trip	56.35	19.27
current_Mpoint	Points promised for this trip	56.52	19.11
flexible	If this user have a flexible travel time	yes=10.5%, no=89.5%	

Previous experience in travel time prediction, prev_exp, is calculated as prev_est_tt-prev_act_tt, as it is suspected that if the previous trips took much longer than expected, the user is more likely to change to another departure time. Also in the model, prev_tt_saving is formulated as a dummy variable as whether prev_tt_saving is larger than certain threshold (say 5 mins). This can explain if the user prefers a certain departure time, or may consider shift departure time only if the saving is beyond expectation. Various combinations of the variables are tested and the results from model explaining the meaningful reasons the commuter shift travel time best is displayed in Table 3.

Three different models are used for testing purpose in this section.

- A *binary logit model* with a dummy dependent variable "same departure time as previous trip", the value of this dummy value is set to be 1 if the new departure time is within 5 minutes from the previous trip, and 0 otherwise.
- A *multinomial logit model* with three alternatives for the dependent variable, as same departure time, leave earlier or leave later.
- *Random effect model*, in which the decision from the same user is expected to have some unobserved correlation.

Testing result of the first binary logit model shows 45.7% of times the user choose the same departure time as previous. The result of the second multinomial logit model shows 45.7% travelers tends to leave at the same time, while 26.5% and 27.8% users will leave earlier or later, respectively. In other words, the result of the second model is similar to the binary logit as there is no significant difference for the

decision as leave earlier or later. As for the third random effect model, the result shows opposite conclusion when compared to the expectation that same users may have some unobserved correlation, and as a result, it is no better than the model traditional binary logit model.

Therefore, the binary logit model is concluded to have the best results, which is shown in Table 3. A positive estimate shows the user is more likely to depart at the same time as the previous trip, and vice versa.

Table 3 Results from binary choice model for departure time shift

Variables	Estimation	Std.Error	t-value	Signif
constant	0.562	0.146	3.856	***
prev_exp within (-2~3 mins)	0.162	0.089	1.809	.
prev_tt_saving > 5 mins	0.215	0.101	2.134	*
prev_act_tt-current_est_tt>3 min	-0.303	0.139	-2.185	*
current_Mpoint-prev_Mpoint	0.006	0.004	1.757	.
flexible	-0.930	0.138	-6.721	***
log-likelihood	-1672.7			
McFadden R^2	0.020			
Likelihood ratio test	chisq = 68.521 (p.value = 2.0813e-13)			

reference level is “same as previous=false”

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’

- ✧ The results in Table 3 indicate there’re in total five elements playing roles in traveler’s departure time decision making process:*App engagement*. The latest experience with the app also plays a role in the departure time choice behavior, when the difference between estimated travel time and the true travel time is within -2~3 min (row “prev_exp within (-2~3 mins)” in table 3), indicating that if the user had positive experience with the smartphone app by receiving accurate traffic information, the user is more likely to leave at the same time. This reveals that user will not shift departure time and stick to the new departure time if the previous experience was good.
- ✧ *Previous travel time saving*. The travel time saving in the previous trip is also significant when user decides to depart at the same time (row “prev_tt_saving > 5 mins” in table 3), evidenced by the positive sign, which means if the previous travel time saving is significant higher than 5 minutes, the driver will have

higher probability of leaving at the same time as last trip. On the contrary, if the saving in travel time is less than 5 minutes the user will learn from the previous travel experience and seek a new departure time with more travel time savings for the next trip.

- *Habitual travel time*. The model results also imply users' decisions are influenced by the information provided by the app, and also this information will be compared with the previous experience. If the estimated travel time for a new departure time is significantly shorter than the previous true travel time (in table 3 we used 3 minutes as the threshold), the user is more likely to change the departure time (with a negative estimates for "prev_act_tt-current_est_tt>3 min"). This also suggested a certain tolerance in the habitual longer travel time, meaning if the travel time increases only slightly, the user won't change their departure time.
- *Incentives*. The impact of rewards on departure time choice is also proved to be effective. The positive sign of the variable "current_Mpoint-prev_Mpoint" in table 3 means the user will choose the same departure time if the rewards promised for this trip is higher than the previous time. The rewards value is related to the estimated travel time. The larger difference in the points means the user may save more travel time and earn more points for the same departure time as previous trip, then the user will prefer to leave at the time he/she has some experience and can also earn more points.
- *Departure time flexibility*: Indicated by the negative sign for the estimate, the result shows that the commuter with flexible hour (row "flexible" in table 3) is less likely to leave for work at exactly same departure time with lastest trip (or within 5 minutes of the previous departure time), i.e. travelers with flexible schedule are more likely to change their departure time. Because this kind of commuters has some flexibility in the arrival time, they may leave to work at certain broader range, and can be either earlier or later.

The result of multinomial logit model is shown in Table 4. Since the reference level of two models is "not same as previous" and "same as previous" respectively, the results have opposite signs.

Table 4 Results from multinomial logit model for departure time shift

Variables	Estimation	Std.Error	t-value	Signif
earlier: constant	-1.257	0.189	-6.638	***
later: constant	-1.258	0.181	-6.928	***
earlier: prev_exp within (-2~3 mins)	-0.271	0.108	-2.504	*
later: prev_exp within (-2~3 mins)	-0.056	0.107	-0.529	
earlier: prev_tt_saving > 5 mins	-0.202	0.121	-1.668	.
later: prev_tt_saving > 5 mins	-0.229	0.122	-1.868	.
earlier: prev_act_tt-current_est_tt>3 min	0.108	0.169	0.639	
later: prev_act_tt-current_est_tt>3 min	0.474	0.158	2.992	**
earlier: current_Mpoint-prev_Mpoint	-0.006	0.004	-1.376	
later: current_Mpoint-prev_Mpoint	-0.007	0.004	-1.567	
earlier: (flexible)	0.980	0.182	5.393	***
later: (flexible)	0.884	0.173	5.110	***
log-likelihood	-2600.7			
McFadden R^2	0.014			
Likelihood ratio test	chisq = 76.376 (p.value = 2.5658e-12)			

reference level is “same”

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

A few observations from this model:

- ✧ *App engagement:* When the difference between estimated travel time and the true travel time is within -2~3 min, the user will have higher probability of leaving at the same time. In contrast, if the difference is large than 3 min or less than -2 min, the user is more likely to leave at different time, especially earlier. This reveals that user will not shift departure time and stick to the previous departure time if the previous experience was good. However, if the previous experience was bad, the users are more likely to leave earlier (coefficient -0.271) than later (coefficient -0.056)
- ✧ *Previous travel time saving:* If the previous travel time saving is higher than 5 minutes, the driver is more likely to leave at the same time at last trip, evidenced by the negative sign. In contrast if lower than 5 minutes, the driver will have higher probability of leaving at earlier or later time. No significant difference is observed for leaving earlier (coefficient -0.202) versus leaving later (coefficient -0.229)
- ✧ *Habitual travel time:* When the estimated travel time for a new departure time is significantly shorter than the previous actual travel time (we used 3 minutes as the threshold), the user is more likely to change the departure time (with positive coefficients for "prev_act_tt-current_est_tt>3 min"), especially later. This means if the current travel time estimation is different from previous actual travel time, users are more likely to leave later (coefficient 0.474, if current_est_tt drops) than leave earlier (coefficient 0.108, if current_est_tt increases)
- ✧ *Incentives:* The negative coefficients of the variable "current_Mpoint-prev_Mpoint" in table mean that the driver will choose the same departure time if the rewards promised for this trip is higher than the previous time. No significant difference is observed for leaving earlier (coefficient -0.006) versus leaving later (coefficient -0.007),_or in other words, the reward points doesn't work differently in motivating earlier or later departure.
- ✧ *Departure time flexibility:* The result shows that the commuter with flexible hour is more likely to change the departure time, either earlier or later, indicated by the positive coefficients. No significant difference is observed for leaving earlier (coefficient 0.980) versus leaving later (coefficient 0.884)

An analysis on the model's prediction accuracy of user behavior is also performed. The entire trip dataset was separated into two groups, 80% data used for training purpose, and the remaining 20% data for validation. The rules for validation are as follow: if the model predicts certain action (i.e. leaving at the same time, or at a different time) and the data exhibit same result, we say the model is making a correct prediction. Otherwise the model makes a false prediction. Among the 3,604 home-to-work trips, after removing some of trips from O-D pairs with limited observations (which make them not suitable for training and validation), about 2,000 trips were selected for training and 500 (496 to be exact) were selected for validation purpose. Among them, the model made correct prediction on 379 trips, or 76.4% out of 496 trips.

5. Summary

In this paper, we study the day-to-day departure time choice behavior of the travelers, using true observation data collected from a smartphone app, "Metropia". From the continuous observation of the user's departure time for the repeated trips, how the decisions evolve from time to time is different depending on the trip purposes. Since the first use of the app, the repeated trips between the same OD starts with a large variation in departure time in the first few trips and then slowly converge to reach a stability as the experiences update. The analysis result in a binary choice model for the shift of departure time proves that users' experience in app engagement, previous travel time saving, habitual travel time, incentives, and commute flexibility are able to trigger day-to-day behavioral change for their morning home-to work commutes. The findings of this research provide insights on the users' adaption to a new traffic information service along with incentives, and behavior changes over a transition period. The outcomes suggest ways to improve ICT services and incentive scheme, and also the transportation operation and demand management.

For the future work, we think firstly a few models being used in this research can be further refined and tested with more users and increasing samples. In addition, besides departure time choice, the other main dimension of behavioral changes is the route choice, which this paper does not touch, but certainly with the continuous observation of traveler's behavior and the supporting data, lots of interesting research can be done in this regard.

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